

2 Content-Centric Cognitive Modeling

The language-endowed intelligent agents (LEIAs) described in this book belong to a family of cognitive systems that aim to emulate the human abilities of perception, reasoning, and action. But LEIA research differs from most cognitive systems research with respect to the following parameters:

1. LEIAs are designed to use large-scale stored knowledge and self-generated situational knowledge to support their reasoning about perception, decision-making, and action.
2. LEIAs are designed to engage in lifelong, humanlike learning, which involves acquiring knowledge by reading and through show-and-tell interactions with humans.
3. LEIAs are prepared to explain their knowledge, reasoning, and operation, with the goal of engendering the trust of their human partners.

It is our emphasis on knowledge content that explains the coinage *content-centric cognitive modeling*.¹

The basic principles of LEIA modeling were listed in chapter 1. This chapter introduces the cognitive architecture underlying LEIAs, outlines their symbolic-empirical hybridization, and describes choices and methodologies underpinning this program of research and development. Comparisons with other approaches are presented in the final section.

2.1 The OntoAgent Cognitive Architecture

Our research group develops LEIAs within a cognitive architecture called OntoAgent. Figure 2.1 presents a high-level sketch. For clarity of presentation, it does not show all processing flows and only hints at the central role played by the agents' knowledge bases throughout their operation.

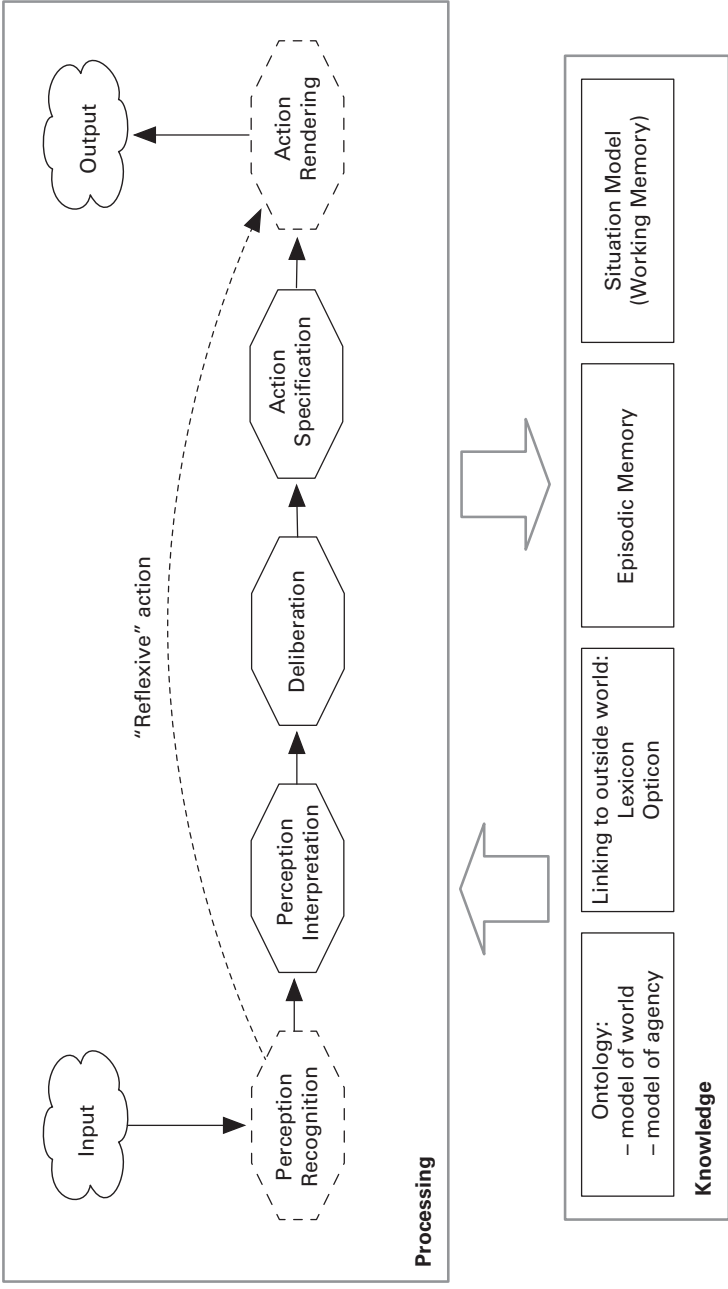


Figure 2.1

A high-level sketch of OntoAgent, the cognitive architecture supporting the development of LEIAs. In the Perception Recognition and Action Rendering modules, data-driven methods play a significant role.

OntoAgent consists of:

1. a set of processing modules—the five octagons in figure 2.1, which are described in sections 2.1.1–2.1.5;
2. a knowledge substrate whose main components are: the ontology; knowledge bases that map elements of the outside world to ontological concepts, such as the lexicon and the opticon; episodic memory; and the situation model, also called working memory; and
3. infrastructure components that support both system functioning (e.g., the code base) and system development (e.g., DEKADE: the Development, Evaluation, Knowledge Acquisition, and Demonstration Environment; cf. section 2.5.2).

The knowledge bases and processing modules are modeled using the ontological metalanguage, which enables metacognition by LEIAs, including the ability to explain their own functioning. Agents process inputs by instantiating the ontological script called AGENT-FUNCTIONING-FLOW, which guides them through the five stages of processing using a sequence of events that they understand, remember, and can explain. This script is described in section 3.2.4, where it is used to illustrate the content and utility of ontological scripts.

In addition to following the five-stage, reasoning-intensive processing flow, agents can also undertake “reflexive” action, shown by the dotted arrow in figure 2.1, which simulates physiological reflexes as well as reasoning by analogy. Reasoning-intensive and reflexive processing have been referred to in the psychological literature as *slow thinking* and *fast thinking*, respectively (see, e.g., Kahneman, 2011).² Choosing between slow and fast thinking is the responsibility of attention management, which scopes over and informs the agent’s basic functioning flow.

The OntoAgent architecture maintains interoperability across its Perception, Deliberation, and Action modules by (a) using a common, ontologically grounded meaning representation language across knowledge bases and internal processors, and (b) making the results of processing available to all modules of the architecture.

To-date, proof-of-concept systems developed on the basis of OntoAgent have demonstrated many functionalities, including language understanding and generation, interpreting the results of visual perception, simulated interoception (i.e., the experiencing of bodily signals), many types of

reasoning and decision-making, learning, simulated action, robotic motor action, and simulated human physiology.

We will now work through the five stages of the processing module of OntoAgent.

2.1.1 Perception Recognition

When a LEIA attends to a new input, it begins with Perception Recognition, which includes two subtasks: detecting the type of input (such as speech, text, or vision) and preparing it for semantic analysis. Perception Recognition incorporates data-driven system modules. The particulars of hybridization, as well as outstanding challenges, differ across perception modalities.

Speech input Speech must be converted into text before it can be semantically analyzed. Speech-to-text conversion is a task for which machine learning is well suited, and good speech-to-text systems are available off the shelf. However, transcription errors are common, particularly if there are overlapping speakers, background noise, fragmentary utterances, or language that contains proper nouns or specialist terminology—a list unsurprising to those familiar with auto-generated closed captioning. In addition, speech-to-text systems do not address prosodic features to the extent needed for full semantic analysis.³ For example, the sequence of words *Tony drove to the store* requires different interpretations depending on intonation and emphasis:

- | | | |
|----------|----------------------------------|---------------------------------------|
| (2.1) a. | Tony drove to the store. | [An assertion.] |
| b. | <i>Tony</i> drove to the store? | [It was Tony, not someone else?] |
| c. | Tony drove to <i>the store</i> ? | [He didn't go somewhere else?] |
| d. | Tony <i>drove</i> to the store? | [He didn't get there some other way?] |
| e. | Tony drove to the store??!! | [I am shocked or outraged.] |

During Perception Recognition, speech-to-text systems need to decorate text segments with linguistically informed prosodic features that can then be interpreted by LEIAs as part of Perception Interpretation. Prosodic feature recognition might well be amenable to machine-learning methods but remains, as yet, understudied (mainstream AI does not pursue a level of semantic or pragmatic analysis that requires it).

Text input Although text might seem like straightforward input to semantic analysis, complexity lurks just beneath the surface. All meaningful typographic conventions need to be semantically or functionally interpreted: italics, boldface, underlining, headings, lists, tables, speaker turns in plays,